

HEAT ILLNESS

HEAT INJURY SEVERITY IS DEPENDENT ON CNS DYSFUNCTION. RECTAL OR OESOPHAGEAL TEMPERATURE IS MOST ACCURATE. MANAGEMENT IS TO ACHIEVE NORMOTHERMIA WITH COOLING AND CRYSTALLOID FLUID CHALLENGE.

FULL PRIMARY SURVEY (ABCDE)

SUSPECTED HEAT-RELATED ILLNESS? CONSIDER OTHER FEBRILE CAUSES

REMOVE FROM HOT ENVIRONMENT AND REMOVE THICK CLOTHING

ASSESSMENT

BREATHING	RATE (<10 VENTILATE), SPO ₂ (94-98%), ETCO ₂ (4.0 to 5.7kPA)		
CIRCULATION	PULSE RATE/STRENGTH (SHOCK), CONTINUOUS ECG, BPs, URINARY OUTPUT		
DISABILITY	AVPU OR GCS, BLOOD SUGAR AND CONTINUOUS RECTAL TEMP MONITORING		
HEAT STRESS	HEAT EXHAUSTION	HEAT STROKE	MULTI-ORGAN
• NORMAL TEMP • HEAT CRAMPS • PERIPHERAL OEDEMA • SWEATING • FLUSHED SKIN • NAUSEA • HEAT SYNCOPE⁽³⁾	• TEMP 37 – 40°C • HEADACHE • CONFUSED • IRRITABLE • TACHYCARDIA • EXCESSIVELY SWEATING • DIARRHOEA AND VOMITING	• REDUCED GCS <13 • TEMP >40°C • HOT, DRY SKIN⁽⁴⁾ • CARDIOVASCULAR COLLAPSE/SHOCK • SEIZURES • LIVER AND RENAL FAILURE • MYOGLOBINURIA	• COMA • ARRYTHMIAS • RESP FAILURE • PERI-ARREST • CARDIAC ARREST

TREATMENT

AIRWAY	P or U = REDUCED AIRWAY REFLEXES - MANAGE WITH ADJUNCTS
BREATHING	MAINTAIN RATE >10, MAINTAIN SPO ₂ 94-98%, MAINTAIN ETCO ₂ (4.0 to 5.7kPA)
CIRCULATION	IF ALERT: ENCOURAGE ORAL FLUIDS WITH ELECTROLYTES IF REDUCED CONSCIOUSNESS: IV ACCESS AND GIVE 250ml BOLUSES OF COLD FLUIDS TO NORMOTENSION ⁽⁵⁾ . CONSIDER IV GLUCOSE 10% IF HYPOGLYCAEMIC
DISABILITY AND COOLING	TARGET 1°C REDUCTION EVERY 15 MINUTES UNTIL MENTAL STATUS IMPROVES HEAT STRESS/EXHAUSTION: REMOVE CLOTHING AND COLD WATER SPRAY HEAT STROKE: PLACE ICE PACKS WRAPPED IN CLOTH INTO NECK, AXILLA, GROIN, PALMS AND SOLES OR IMMERSE IN COLD/ICE WATER UNTIL 39°C SEIZURES: GIVE 10mg MIDAZOLAM IV or 10mg DIAZEPAM IV FOR ADULTS >50kg

HAS MENTAL STATUS IMPROVED WHEN TEMPERATURE IS <39°C?

NO

CONSIDER OTHER DIAGNOSES?
(SEPSIS, DRUG OVERDOSE)

CONVEY TO NEAREST MEDICAL
FACILITY WITH ICU CAPABILITY

YES

REDUCE COOLING MEASURES TO
AVOID OVERTHROW HYPOTHERMIA⁽⁶⁾

CONVEY TO NEAREST MED FACILITY
WITH CONTINUOUS MONITORING

POINTS TO NOTE

1. CNS DYSFUNCTION IS THE BIGGEST INDICATOR OF SEVERITY, ESPECIALLY AS TYMPANIC TEMPERATURE MONITORS CAN BE INACCURATE. CLINICAL JUDGEMENT SHOULD INCLUDE ENVIRONMENTAL FACTORS.
2. CHECK TEMPERATURE AND BLOOD PRESSURE EVERY 10-15 MINUTES IN THE FIRST TWO HOURS.
3. HEAT SYNCOPE OCCURS MAINLY IN PATIENTS THAT HAVE DONE EXERTIONAL EXERCISE IN HOT CLIMATES CAUSING SIGNIFICANT VASODILATION. VENOUS POOLING OCCURS RESULTING IN REDUCED VENOUS RETURN AND ULTIMATELY COLLAPSE. MANAGEMENT INVOLVES TRENDELENBURG POSITIONING AND ENCOURAGEMENT OF ORAL FLUIDS/ELECTROLYTES WHEN MENTAL STATUS IMPROVES. IF MENTAL STATUS DOES NOT IMPROVE AFTER SIMPLE MEASURES THEN TREAT AS WORSENING HEAT ILLNESS.
4. ANHYDROSIS (STOPPED SWEATING) IS A VERY LATE SIGN OF HEAT STROKE.
5. MOST HEAT STROKE PATIENTS WILL REQUIRE 1L OF SALINE WITHIN THE FIRST HOUR.
6. IMMERSION IN COLD/ICE WATER SHOULD BE STOPPED WHEN THE CORE TEMPERATURE REACHES AROUND 39°C TO AVOID OVERSHOOT HYPOTHERMIA. DO NOT IMMERSE THE PATIENT BEYOND THE COSTAL MARGIN.
7. RHABDOMYOLYSIS AND ACUTE KIDNEY INJURY IS A CONCERN WITH HEAT STROKE PATIENTS SO FLUID RESUSCITATION IS FIRST LINE TREATMENT.
8. IF MYOGLOBINURIA OCCURS, AGGRESSIVELY TREAT WITH IV CRYSTALLOID FLUIDS AT 1-2ml/kg/hr, ADMINISTERING 50ml SODIUM BICARBONATE 8.4% WITH EVERY 4th BAG OF FLUID. IF NORMOTENSIVE ALTERNATE CRYSTALLOID FLUID WITH 5% GLUCOSE.
9. ASSESS FOR HYPERKALAEMIA IN THESE PATIENTS AND TREAT WITH CALCIUM CHLORIDE 10% IF ECG SHOWS ABNORMALITIES SUCH AS PEAKED T WAVES, FLATTENING P WAVES AND WIDE QRS COMPLEXES.
10. AVOID PARACETAMOL AND IBUPROFEN IN THESE PATIENTS AS IT MAY CAUSE LIVER/RENAL DYSFUNCTION.
11. OTHER DIAGNOSES COULD INCLUDE SEPSIS OR TOXIC SUBSTANCE OVERDOSES. REVIEW POISONING AND TOXIC INJURIES GUIDELINE.

Guideline written in accordance with SFM Clinical Standard Operating Procedure 012 'Management of Heat Illness' (2022) and MERT Standard Operating Procedure 'Heat Illness V3.0'

References

- Epstein, Y. and Yanovich, R. (2019) Heatstroke, *New England Journal of Medicine*, 380(25), pp. 2449-2459.
- Gauer, R. and Meyers, B. K. (2019) Heat-related Illnesses, *Am Fam Physician*, 99(8), pp. 482-489.
- Lipman, G. S., et al. (2019) Wilderness Medical Society Clinical Practice Guidelines for the Prevention and Treatment of Heat Illness: 2019 Update, *Wilderness Medical Society Clinical Practice Guidelines*, 30(4), pp. S33-S46.
- NICE [NG148] (2019). *Acute Kidney Injury: Prevention, Detection and Management*. Available at: [Overview | Acute kidney injury: prevention, detection and management | Guidance | NICE](#) (Accessed 19 March 2023).